Assignment #1

- Cyclic Rotation

```
FUNCTION solutions(A, K)
    n = length of array A
    Initialize b as an empty array with length n

    FOR every element in A
        new position = (index of element + K) MOD n
        b(new position) = element

    RETURN b
```

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- Frog Jump

```
FUNCTION solution(X, Y, D)
    jumps = 0
    distance = Y - X
    WHILE distance > 0
        decrement distance with D
        increment jumps by 1
    RETURN jumps
```

- Odd Occurences In Array

```
FUNCTION solution(A)
    FOR the length of the array
        IF this element has an
            odd number of instances
    RETURN this elemt
```

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- Missing Integer

```
FUNCTION solution(A)
    min index = 0
    FOR every elemnt in the array
        IF A(min index) > this elemnt
            min index = this index
    n = 1
    IF A(min index) ≤ 0 AND n NOT IN A
        RETURN n
    ELIF A(min index) > n
        RETURN n
    ELSE
        WHILE n IN A
            n += 1
        RETURN n
```

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Max Counters

```
FUNCTION solution(N, A)
    Initialize array counters w/ length N w/ 0
    min = 0
    max = 0
    FOR every elmnt in A
        IF this elmnt is b/w 1 & N
            IF the counter of this elmnt < min
                add the current min + 1
            ELSE
                increment the counter
        IF the counter of this elmnt > min
            Max = the counter of this elmnt
```

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ELIF this element = N+1

min = max

FOR every element in counters

IF this counters < min

set this counter = min

RETURN counters

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